Does PoPE's pinned phase offset explain the MapPoPE interaction? -- results (registered, `DELTA_FREE_PREREG.md` + Amendment 1)

Batch 2026-10-20 10:26-14:00 on GPU 1 (shared with another user, <= 4 runs), 40 runs, seeds 1600-1607 (fresh). Paper torus task, T = 128, rank 2, PAPER2X2's recipe (300 epochs, cosine, lr 1e-3, wd 0.05). Arms: PoPE-Flat (index phase, delta pinned: zero init held at 0 by the clamp's dead zone), PoPE-FreeDelta (delta init U(-2 pi, 0), clamp removed), MapPoPE-Flat (path phase, pinned), MapPoPE-FreeDelta (path, free), MapPoPE-FrozenDelta (path, delta frozen at the free arms' draw). At each seed twins differ only in delta. Independent check before launch (`docs/audits/2026-10-20/delta_free_audit.md`, Amendment 1); pilot on seed 1690 (never pooled). Analysis `analyze_delta_free.py` -> `DELTA_FREE_ANALYSIS.txt`, `DELTA_FREE_VERDICTS.json`; recipe asserted on all 40 checkpoints. `experiment_audit`: all arms converged, windowed classes agree. Own-stream retrace floor 0.938. Account under test (`docs/paper/expansion/MAPPOPE.md`): PoPE's content-free offset delta, pinned at 0, fixes every kernel's peak at zero phase difference; with a path phase that is the revisit target (helps), with an index phase it is the query's own token (hurts: no multi-step retrace kernel).

PRIMARY (registered): PEAK-PINNING SUPPORTED (path unchanged within noise). Freeing delta raises the index model by +0.122 [90% CI +0.076, +0.167] (8/8 seeds, sign-flip p 0.0078), concentrated in multi-step reversals (R3+ +0.180, Holm p 0.031), and leaves the path model at ceiling (-0.018, p 0.31; SOLVED 6/8 vs 8/8, Fisher p 0.47).

arm	T = 128	T = 512	T = 1024	SOLVED
PoPE-Flat (index, pinned)	0.649	0.615	0.599	0/8
PoPE-FreeDelta (index, free)	**0.771**	0.536	0.442	0/8
MapPoPE-Flat (path, pinned)	0.999	0.980	0.933	8/8
MapPoPE-FreeDelta (path, free)	0.981	0.901	0.831	6/8
MapPoPE-FrozenDelta (path, random fixed)	0.999	0.929	0.832	8/8

Interaction I = dP - dI = -0.140 [-0.186, -0.092] (0/8 seeds positive, p 0.0078); PAPER2X2's pinned score x phase interaction was +0.154 (cross-batch, descriptive). The licensed sentence (Amendment 1 MAJOR 5) is the one above, not "freeing delta reverses the interaction".

Mechanism readouts (declared secondaries; eval-only probes, rebuilt logits exact)

- **Index kernel:** with delta pinned, 0% of content pairs prefer offset 3 over offset 1 (kernel argmax at offset 1 in 1074/1088 pairs); freed, 50-100% per seed, with argmax spread over offsets 3, 4, 7, 8 (the retrace comb). Attention on the newest observation (never the target) 0.287 -> 0.003.
- **Where the index gain comes from:** R1 +0.039 (n.s.), R2 +0.117, R3+ +0.180 (Holm 0.031 each), unvisited-run +0.014; R3+ carries 0.093 of the 0.122.

- **Path target alignment:** C (same-cell key alignment) 0.987 pinned vs 0.616 free and 0.614 frozen; same-cell attention mass 0.909 vs 0.809 / 0.876. Accuracy at T = 128 stays at ceiling because the code itself (C0 0.993 / 0.798 / 0.884) still carries the map.
- **Delta (S4):** pinned arms 96-100% in the dead zone. Free arms moved mostly by weight decay (x0.447); the learned part beyond decay is largest in the fastest frequency quartile (index 0.447, path 0.371 rad).
- **S7 (diagnostic):** resetting delta to 0 on either half of the free index model's channels drops it below the pinned model (0.43 / 0.47 vs 0.649): the trained network relies on its learned offsets across the spectrum, so the reset is not a clean ablation of one mechanism.

Beyond the training length (robustness, rule 10)

- **P1 (registered):** path frozen / free minus pinned at T = 1024: -0.101 (p 0.047, Holm 0.094) and -0.102 (Holm 0.195): both UNMEASURED by the registered rule; descriptively both lose robustness (T = 512: -0.052, -0.079).
- Index free minus pinned at T = 1024: -0.158 (p 0.016, Holm 0.094): the retrace kernel the free offset buys is length-specific, while the pinned recency kernel extrapolates (descriptive).
- Training (S6): the free index arm learns faster (loss AUC -0.131, p 0.0078); the free and frozen path arms slower (+0.122, +0.040).

Reading

PoPE's polar score helps a path-integrated phase and hurts an index phase because its pinned offset fixes the attention kernel's peak at zero phase difference. Freeing the offset lets the index model build the multi-step retrace kernel RoPE uses (+0.122 at T = 128, mostly on multi-step reversals) and leaves the path model's accuracy at ceiling while its target alignment falls (0.987 -> 0.62). The account is causal at T = 128; the costs beyond the training length are descriptive. Scope: one layer, 2 heads, rank 2, the paper torus at T = 128.